

Project 7

Mark Wang

Contents

(a)	1
(b) 1	3
(b) 2	4
(b) 3	5
(b) 4	5
Appendix: full code	6

Portfolios are built on the training period, the same 59 returns used in Projects 1, 2, 4, 5 and 6, with $R_f = 0.001$. Part (b) then evaluates them on the 54 months of the testing period, with the weights held fixed at their training period values.

(a)

The multigroup model assumes correlation is constant within each group and constant between any two groups, so the 30×30 correlation matrix collapses to the 5×5 matrix of the five sectors:

Table 1: Within and between group correlations

	Technology	Communication	Financial	Consumer Cyclical	Healthcare
Technology	0.4363	0.3277	0.3464	0.3567	0.2532
Communication	0.3277	0.3263	0.4395	0.3282	0.2185
Financial	0.3464	0.4395	0.6909	0.3555	0.2679
Consumer Cyclical	0.3567	0.3282	0.3555	0.3349	0.2530
Healthcare	0.2532	0.2185	0.2679	0.2530	0.3786

For stock i in group k ,

$$z_i = \frac{1}{(1 - \rho_{kk})\sigma_i} \left(\frac{\bar{R}_i - R_f}{\sigma_i} - C_k \right)$$

where the group cut-offs solve the $K \times K$ system of handout #37,

$$C_k + \sum_{l=1}^K \frac{\rho_{kl}N_l}{1 - \rho_{ll}} C_l = \sum_{l=1}^K \frac{\rho_{kl}A_l}{1 - \rho_{ll}}, \quad A_l = \sum_{i \in l} \frac{\bar{R}_i - R_f}{\sigma_i}$$

Table 2: Group sums and cut-offs

Group	A_k	C_k
Technology	2.5125	0.338994
Communication	1.0441	0.233058
Financial	1.2471	0.207766
Consumer Cyclical	1.7458	0.266283
Healthcare	1.6375	0.247979

```
#> max |handout 37 route - Sigma^-1 (Rbar - Rf 1)| = 2.11e-14
```

Table 3: Composition of the multigroup optimal portfolio, short sales allowed

Stock	Group	Weight
NVDA	Technology	0.0542
AAPL	Technology	0.1211
MSFT	Technology	0.7741
AVGO	Technology	0.0121
AMD	Technology	0.0153
NOW	Technology	0.3231
GOOG	Communication	0.2851
META	Communication	-0.0172
NFLX	Communication	0.0931
VZ	Communication	-0.4443
DIS	Communication	-0.2348
T	Communication	-0.7587
V	Financial	0.4340
JPM	Financial	-0.0112
MS	Financial	0.0578
GS	Financial	-0.2214
BAC	Financial	-0.0691
SCHW	Financial	-0.0753
AMZN	Consumer Cyclical	0.1090
TSLA	Consumer Cyclical	0.0610
MCD	Consumer Cyclical	0.1452
SBUX	Consumer Cyclical	-0.0734
HD	Consumer Cyclical	0.1942
EBAY	Consumer Cyclical	-0.1480
LLY	Healthcare	0.2315
NVO	Healthcare	0.4219
UNH	Healthcare	0.2015
JNJ	Healthcare	-0.1895
AMGN	Healthcare	-0.2475
PFE	Healthcare	-0.0442

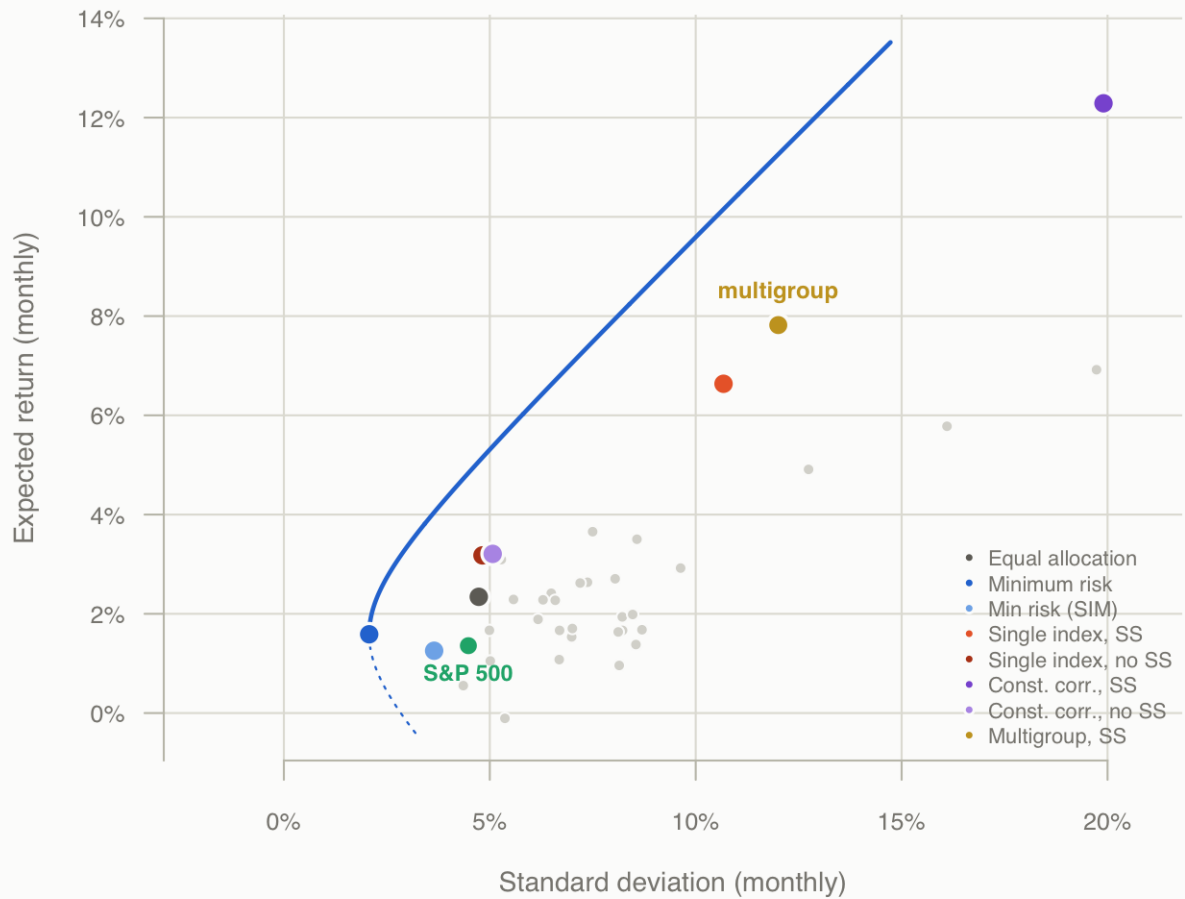
```
#> E = 7.8187% per month
```

```
#> sd = 12.0046% per month
```

```
#> 13 short positions, sum|x| = 6.07
```

(a) The multigroup portfolio alongside every earlier portfolio

Training period, historical covariance matrix. Blue curve is Project 2's frontier.



(b) 1

Growth of one dollar over the testing period, weights fixed:

(b1) Performance over the testing period

Feb 2022 to Jul 2026, weights fixed at their training period values.

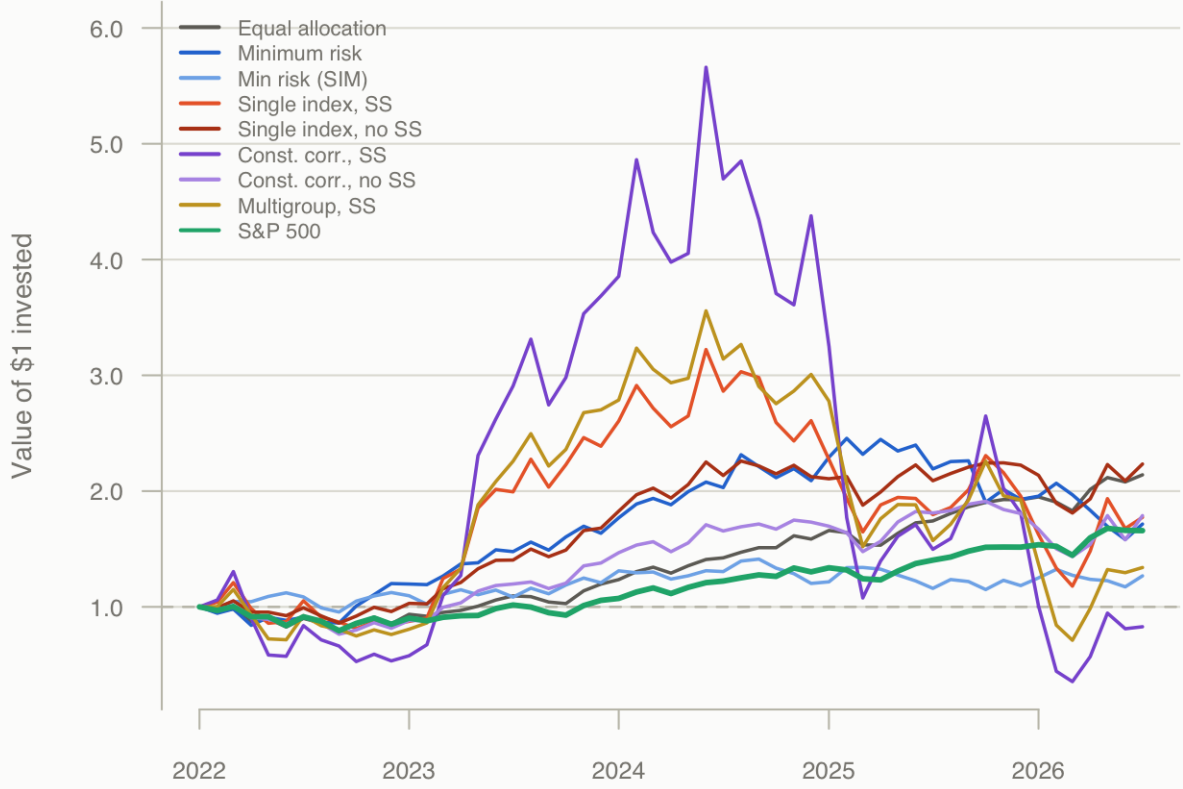


Table 4: Value at 31-Jul-2026 of one dollar invested at the start

Portfolio	Final value of \$1
Equal allocation	2.1404
Minimum risk	1.7153
Min risk (SIM)	1.2677
Single index, SS	1.7736
Single index, no SS	2.2338
Const. corr., SS	0.8283
Const. corr., no SS	1.7880
Multigroup, SS	1.3399
S&P 500	1.6587

(b) 2

Average growth as the geometric mean, $(\prod_t (1 + R_t))^{1/T} - 1$:

Table 5: Monthly average growth over the testing period

Portfolio	Geometric (%)	Arithmetic (%)	Std. dev. (%)
Equal allocation	1.4193	1.5351	4.8569
Minimum risk	1.0042	1.2280	6.7479
Min risk (SIM)	0.4402	0.5613	4.9952
Single index, SS	1.0668	1.9359	13.7472
Single index, no SS	1.4995	1.6882	6.2247
Const. corr., SS	-0.3481	3.5439	28.4954
Const. corr., no SS	1.0819	1.2950	6.6234
Multigroup, SS	0.5433	1.9156	16.7019
S&P 500	0.9414	1.0440	4.5709

(b) 3

$$S_p = \frac{\bar{R}_p - R_f}{\sigma_p}, \quad T_p = \frac{\bar{R}_p - R_f}{\beta_p}, \quad \alpha_p = \bar{R}_p - [R_f + \beta_p(\bar{R}_m - R_f)]$$

and the differential excess return, the gap to the point on the capital market line carrying the same total risk,

$$\bar{R}_p - \left[R_f + \frac{\sigma_p}{\sigma_m}(\bar{R}_m - R_f) \right]$$

Table 6: Performance measures, testing period. Betas estimated on the testing period.

Portfolio	beta	Sharpe	Diff. excess (%)	Treynor	Jensen (%)
Equal allocation	1.0315	0.2955	0.4320	0.0139	0.4613
Minimum risk	0.4387	0.1672	-0.2656	0.0257	0.7139
Min risk (SIM)	0.1369	0.0923	-0.5703	0.0337	0.3321
Single index, SS	1.2892	0.1335	-1.0032	0.0142	0.6189
Single index, no SS	0.9498	0.2551	0.3027	0.0167	0.6917
Const. corr., SS	2.7244	0.1209	-2.4410	0.0126	0.8721
Const. corr., no SS	1.0345	0.1804	-0.1729	0.0116	0.2184
Multigroup, SS	1.7586	0.1087	-1.6337	0.0103	0.1555
S&P 500	1.0000	0.2065	0.0000	0.0094	0.0000

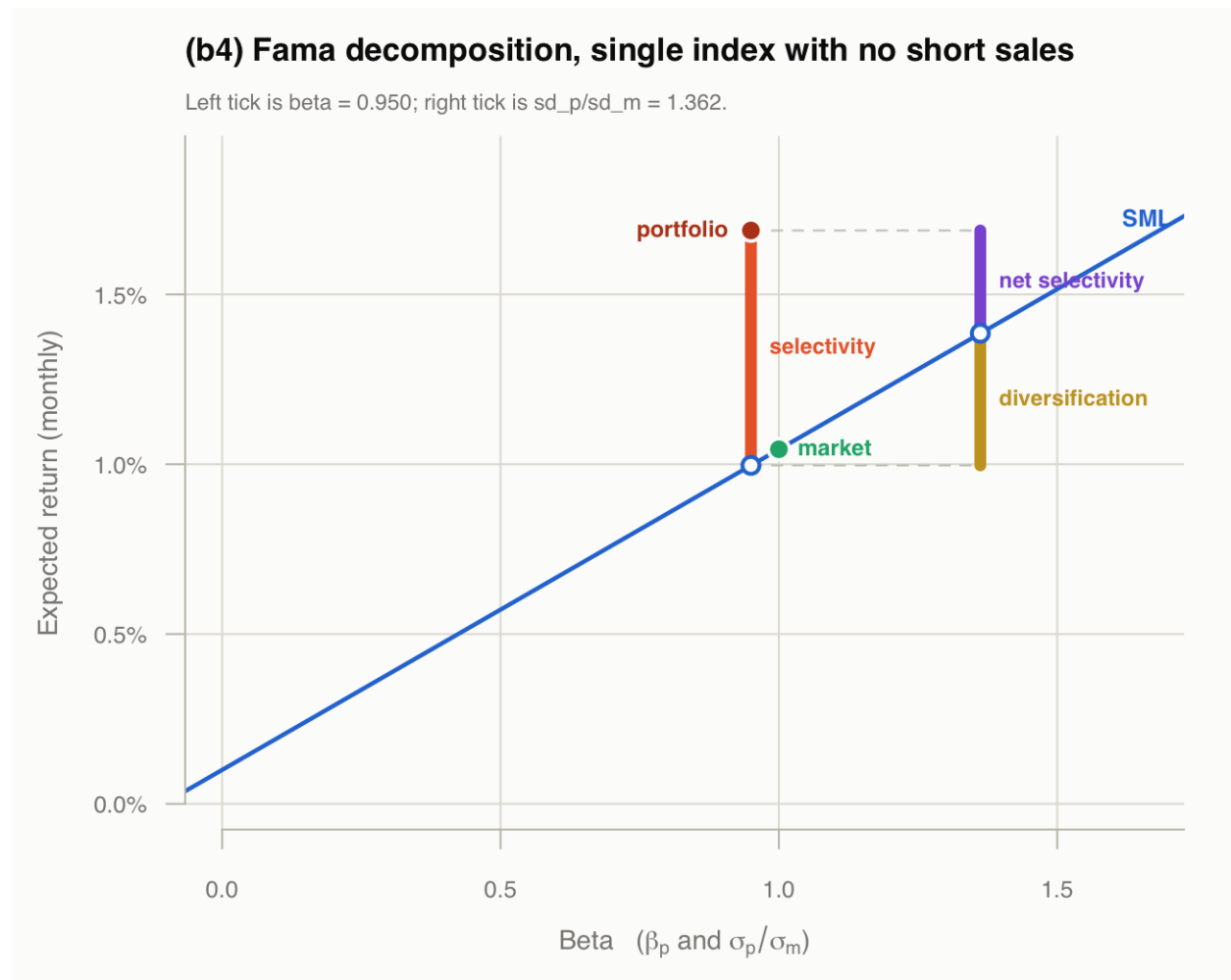
(b) 4

Fama's decomposition of Single index, no SS, splitting the overall gain over R_f into the part earned by taking systematic risk and the part earned by selection, then splitting selection again by charging the portfolio for the diversification it gave up:

$$\underbrace{\bar{R}_p - R_f}_{\text{overall}} = \underbrace{\beta_p(\bar{R}_m - R_f)}_{\text{risk}} + \underbrace{\left[\frac{\sigma_p}{\sigma_m} - \beta_p \right] (\bar{R}_m - R_f)}_{\text{diversification}} + \underbrace{\bar{R}_p - R_f - \frac{\sigma_p}{\sigma_m}(\bar{R}_m - R_f)}_{\text{net selectivity}}$$

Component	Percent per month
Overall performance	1.5882
from systematic risk	0.8966
from selectivity	0.6917
diversification	0.3890
net selectivity	0.3027

```
#> beta = 0.9498
#> sd_p / sd_m = 1.3618
#> risk + selectivity - overall = 0.00e+00
#> diversification + net selectivity - sel = 0.00e+00
```



The portfolio sits at $\beta_p = 0.950$, and its total risk corresponds to a beta of $\sigma_p/\sigma_m = 1.362$ had it been fully diversified. Diversification is the SML distance between those two betas; net selectivity is what is left over the SML at the higher one.

Appendix: full code

```
# Project 7: the multigroup model, and out of sample evaluation of every
# portfolio built in Projects 1 to 7.
```

```

local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "performance.R")
    setwd(dirname(normalizePath(f)))
})

DATA <- ".."

# ---- data -----
# Training period builds the portfolios, testing period evaluates them.

read_prices <- function(f) {
  a <- read.csv(file.path(DATA, f), sep = ",", header = TRUE, check.names = FALSE)
  stopifnot(!any(duplicated(names(a))))
  list(P = as.matrix(a[, -1][, -1]), d = as.Date(a$date))
}

tr <- read_prices("stockData_train.csv")
te <- read_prices("stockData_test.csv")

rets <- function(P) { n <- nrow(P); P[-1, ] / P[-n, ] - 1 }
ret1 <- rets(tr$P) # 59 x 31
ret2 <- rets(te$P) # 54 x 31
dat2 <- te$d[-1]
stopifnot(nrow(ret1) == 59, nrow(ret2) == 54)

stk <- setdiff(colnames(ret1), "^GSPC")
r1 <- ret1[, stk]; r2 <- ret2[, stk]
Rm1 <- as.numeric(ret1[, "^GSPC"]); Rm2 <- as.numeric(ret2[, "^GSPC"])
N <- length(stk)
stopifnot(N == 30)

Rbar <- colMeans(r1)
sdv <- apply(r1, 2, sd)
S_hist <- cov(r1)
Rf <- 0.001 # same Rf as Projects 5 and 6

# single index inputs
fit <- lapply(stk, function(s) lm(r1[, s] ~ Rm1))
names(fit) <- stk
beta <- vapply(fit, function(f) unname(coef(f)[2]), numeric(1))
sig2e <- vapply(fit, function(f) summary(f)$sigma^2, numeric(1))
sig2m <- var(Rm1)
S_sim <- outer(beta, beta) * sig2m
diag(S_sim) <- beta^2 * sig2m + sig2e
dimnames(S_sim) <- list(stk, stk)

# Project 2's frontier, for the plot
one <- rep(1, N)
Si <- solve(S_hist)
fA <- as.numeric(t(one) %*% Si %*% Rbar); fB <- as.numeric(t(Rbar) %*% Si %*% Rbar)
fC <- as.numeric(t(one) %*% Si %*% one); fD <- fB * fC - fA^2
stopifnot(abs(fA - 36.9864) < 1e-3, abs(fC - 2330.03) < 1e-1)

```

```

sd_at <- function(E) sqrt(1 / fC + (fC / fD) * (E - fA / fC)^2)

# ---- (a) the multigroup model -----
# Five sectors of six stocks each, in the order the data files are laid out.
# Correlation is assumed constant within a group and constant between any two
# groups, so the 30 x 30 correlation matrix collapses to a 5 x 5 one.

grp      <- rep(1:5, each = 6)
gnames   <- c("Technology", "Communication", "Financial", "Consumer Cyclical",
              "Healthcare")
names(grp) <- stk
K <- 5; Nk <- as.numeric(table(grp))
stopifnot(all(Nk == 6))

CCm <- cor(r1)
rho <- matrix(0, K, K, dimnames = list(gnames, gnames))
for (k in 1:K) for (l in 1:K) {
  blk <- CCm[grp == k, grp == l]
  rho[k, l] <- if (k == l) mean(blk[upper.tri(blk)]) else mean(blk)
}
stopifnot(isSymmetric(rho), all(diag(rho) > 0))

# the multigroup covariance matrix
S_mg <- rho[grp, grp] * outer(sdv, sdv)
diag(S_mg) <- sdv^2
dimnames(S_mg) <- list(stk, stk)
stopifnot(isSymmetric(S_mg), min(eigen(S_mg, only.values = TRUE)$values) > 0)

# Solved directly,  $Z = \Sigma^{-1} (\bar{R} - R_f \mathbf{1})$ 
Z_mg <- solve(S_mg, Rbar - Rf)
x_mg <- Z_mg / sum(Z_mg)

# Handout #37 solves it through the group cut-offs instead. With
#  $A_k = \text{sum of } (\bar{R}_{i,k} - R_f) / \text{sd}_i \text{ over group } k$ , the  $C_k$  satisfy
#  $C_k + \text{sum}_l [\rho_{kl} N_l / (1 - \rho_{ll})] C_l = \text{sum}_l \rho_{kl} A_l / (1 - \rho_{ll})$ 
# and then  $z_i = [(\bar{R}_i - R_f) / \text{sd}_i - C_k] / ((1 - \rho_{kk}) \text{sd}_i)$ .
ratio <- (Rbar - Rf) / sdv
Ak      <- vapply(1:K, function(k) sum(ratio[grp == k]), numeric(1))
M       <- outer(rep(1, K), Nk / (1 - diag(rho))) * rho
bvec    <- as.numeric(rho %*% (Ak / (1 - diag(rho))))
Ck      <- solve(diag(K) + M, bvec)
names(Ck) <- gnames

z_h37 <- (ratio - Ck[grp]) / ((1 - diag(rho)[grp]) * sdv)
stopifnot(max(abs(z_h37 - Z_mg)) < 1e-9)      # the two routes agree
stopifnot(abs(sum(x_mg) - 1) < 1e-12)

cat("(a) multigroup model\n")
cat("    correlation within and between groups:\n")
print(round(rho, 4))
cat("\n    group cut-offs C_k:\n")
print(round(Ck, 6))
cat(sprintf("\n    max |handout 37 route - direct solve| = %.2e\n",

```



```

      max(abs(z_h37 - Z_mg))))
cat(sprintf("      %d short positions, sum|x| = %.2f\n\n", sum(x_mg < 0), sum(abs(x_mg))))

# ---- the portfolios from every project -----

# Rebuilt exactly as in Project 6: rank, find the cut-off, weight.

egp_sim <- function(short) {
  ratio <- (Rbar - Rf) / beta
  ord <- order(ratio, decreasing = TRUE)
  num <- (Rbar - Rf) * beta / sig2e
  den <- beta^2 / sig2e
  Ci <- sig2m * cumsum(num[ord]) / (1 + sig2m * cumsum(den[ord]))
  k <- if (short) N else max(which(ratio[ord] > Ci))
  inc <- stk[ord][seq_len(k)]
  z <- (beta[inc] / sig2e[inc]) * (ratio[inc] - Ci[k])
  z / sum(z)
}

rho_c <- mean(CCm[upper.tri(CCm)])

egp_cc <- function(short) {
  ratio <- (Rbar - Rf) / sdv
  ord <- order(ratio, decreasing = TRUE)
  Ci <- (rho_c / (1 - rho_c + seq_len(N) * rho_c)) * cumsum(ratio[ord])
  k <- if (short) N else max(which(ratio[ord] > Ci))
  inc <- stk[ord][seq_len(k)]
  z <- (1 / ((1 - rho_c) * sdv[inc])) * (ratio[inc] - Ci[k])
  z / sum(z)
}

x_sim_ss <- egp_sim(TRUE); x_sim_noss <- egp_sim(FALSE)
x_cc_ss <- egp_cc(TRUE); x_cc_noss <- egp_cc(FALSE)
stopifnot(all(x_sim_noss > 0), all(x_cc_noss > 0))

pad <- function(x) { w <- setNames(rep(0, N), stk); w[names(x)] <- x; w }

port <- list(
  "Equal allocation" = setNames(rep(1 / N, N), stk),
  "Minimum risk" = as.numeric(Si %*% one / fC),
  "Min risk (SIM)" = as.numeric(solve(S_sim, one) / sum(solve(S_sim, one))),
  "Single index, SS" = pad(x_sim_ss),
  "Single index, no SS" = pad(x_sim_noss),
  "Const. corr., SS" = pad(x_cc_ss),
  "Const. corr., no SS" = pad(x_cc_noss),
  "Multigroup, SS" = pad(x_mg)
)
port <- lapply(port, function(w) setNames(as.numeric(w), stk))
stopifnot(all(abs(vapply(port, sum, numeric(1)) - 1) < 1e-9))

# the short sales single index portfolio must still be Project 5 part (a)
Z5 <- solve(S_sim, Rbar - Rf)
stopifnot(max(abs(port[["Single index, SS"]] - Z5 / sum(Z5))) < 1e-9)

```

```

# training period statistics, for the plot
tr_E <- vapply(port, function(w) sum(w * Rbar), numeric(1))
tr_sd <- vapply(port, function(w) sqrt(as.numeric(t(w) %*% S_hist %*% w)), numeric(1))

cat("(a) the multigroup portfolio on the training period\n")
cat(sprintf("      E = %.4f%%      sd = %.4f%% (historical covariance matrix)\n\n",
            tr_E["Multigroup, SS"] * 100, tr_sd["Multigroup, SS"] * 100))

# ---- (b) out of sample evaluation, 01-Jan-2022 to 31-Jul-2026 -----

pr <- vapply(port, function(w) as.numeric(r2 %*% w), numeric(nrow(r2)))
pr <- cbind(pr, "S&P 500" = Rm2)
labels <- colnames(pr)

# b1: growth of one dollar
growth <- rbind(1, apply(1 + pr, 2, cumprod))

# b2: geometric mean
geo <- apply(1 + pr, 2, function(z) prod(z)^(1 / length(z)) - 1)

# b3: Sharpe, differential excess return, Treynor, Jensen
mu <- colMeans(pr)
sdp <- apply(pr, 2, sd)
bet <- apply(pr, 2, function(z) cov(z, Rm2) / var(Rm2))
mu_m <- mean(Rm2); sd_m <- sd(Rm2)

sharpe <- (mu - Rf) / sdp
treynor <- (mu - Rf) / bet
jensen <- mu - (Rf + bet * (mu_m - Rf))
diffex <- mu - (Rf + (sdp / sd_m) * (mu_m - Rf))

stopifnot(abs(sharpe["S&P 500"] - (mu_m - Rf) / sd_m) < 1e-12)
stopifnot(abs(jensen["S&P 500"]) < 1e-12, abs(diffex["S&P 500"]) < 1e-12)

perf <- data.frame(portfolio = labels, geometric = geo, mean = mu, sd = sdp,
                  beta = bet, sharpe = sharpe, diff_excess = diffex,
                  treynor = treynor, jensen = jensen, row.names = NULL)

cat("(b) testing period", format(dat2[1]), "to", format(dat2[length(dat2)]),
    "-", nrow(pr), "months\n\n")
print(within(perf, {
  geometric <- round(geometric * 100, 4); mean <- round(mean * 100, 4)
  sd <- round(sd * 100, 4); beta <- round(beta, 4); sharpe <- round(sharpe, 4)
  diff_excess <- round(diff_excess * 100, 4); treynor <- round(treynor, 4)
  jensen <- round(jensen * 100, 4)
}))
cat("\n      (geometric, mean, sd, diff_excess and jensen in percent per month)\n\n")

# b4: Fama decomposition for the single index portfolio with no short sales
fp <- "Single index, no SS"
f_mu <- mu[fp]; f_sd <- sdp[fp]; f_beta <- bet[fp]

fama <- c(

```

```

overall      = unname(f_mu - Rf),
risk         = unname(f_beta * (mu_m - Rf)),
selectivity  = unname(f_mu - Rf - f_beta * (mu_m - Rf)),
diversif     = unname((f_sd / sd_m - f_beta) * (mu_m - Rf)),
net_select   = unname(f_mu - Rf - (f_sd / sd_m) * (mu_m - Rf))
)
stopifnot(abs(fama["overall"] - (fama["risk"] + fama["selectivity"])) < 1e-15)
stopifnot(abs(fama["selectivity"] - (fama["diversif"] + fama["net_select"])) < 1e-15)
stopifnot(abs(fama["selectivity"] - jensen[fp]) < 1e-15)

beta_eq <- unname(f_sd / sd_m)           # beta a fully diversified portfolio
                                         # with this total risk would carry

cat("(b4) Fama decomposition,", fp, "\n")
cat(sprintf("      beta %.4f, sd %.4f%%, beta equivalent of total risk %.4f\n",
            f_beta, f_sd * 100, beta_eq))
cat(sprintf("      overall performance  %8.4f%%\n", fama["overall"] * 100))
cat(sprintf("      from systematic risk %8.4f%%\n", fama["risk"] * 100))
cat(sprintf("      from selectivity      %8.4f%%\n", fama["selectivity"] * 100))
cat(sprintf("      diversification      %8.4f%%\n", fama["diversif"] * 100))
cat(sprintf("      net selectivity      %8.4f%%\n\n", fama["net_select"] * 100))
# ---- plotting helpers -----

surface <- "#fcfcfb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_front <- "#2a78d6"; c_eq <- "#eb6834"; c_spx <- "#1baf7a"; c_arb <- "#6f6d67"

open_png <- function(f, h = 4.7) {
  png(f, width = 6.5, height = h, units = "in", res = 200)
  par(bg = surface, mar = c(4.2, 4.6, 3.4, 1.4), family = "sans")
}
ax <- function(side, at, lab) {
  axis(side, at = at, labels = lab, col = "#c3c2b7", col.axis = muted,
       cex.axis = 0.75, lwd = 1, las = if (side == 2) 1 else 0)
}
pt <- function(x, y, col, lab, pos = 4, cex = 0.72, off = 0.6) {
  points(x, y, pch = 21, bg = col, col = surface, lwd = 1.8, cex = 1.5)
  text(x, y, lab, pos = pos, offset = off, cex = cex, font = 2, col = col)
}

pal <- c("Equal allocation"      = "#6f6d67", "Minimum risk"      = "#2a78d6",
        "Min risk (SIM)"        = "#7fb2ea", "Single index, SS"   = "#eb6834",
        "Single index, no SS"   = "#b8431a", "Const. corr., SS"   = "#8a5cd6",
        "Const. corr., no SS"   = "#b79ae8", "Multigroup, SS"    = "#c9a227",
        "S&P 500"              = "#1baf7a")

# ---- (a) every portfolio on the training period axes -----

open_png("fig_multigroup.png", h = 5.4)
xr <- c(-0.02, max(c(sdv, tr_sd)) * 1.05)
yr <- c(-0.004, max(c(Rbar, tr_E)) * 1.10)
xtick <- pretty(c(0, max(c(sdv, tr_sd))))
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = xtick, col = grid, lwd = 1)

```

```

Eg <- seq(yr[1], yr[2], length.out = 800); sdg <- sd_at(Eg)
eff <- Eg >= fA / fC
lines(sdg[!eff], Eg[!eff], col = c_front, lwd = 1.1, lty = 3)
lines(sdg[ eff], Eg[ eff], col = c_front, lwd = 2.2)

points(sdv, Rbar, pch = 21, bg = "#d9d8d2", col = surface, lwd = 1, cex = 0.8)
points(sd(Rm1), mean(Rm1), pch = 21, bg = c_spx, col = surface, lwd = 1.6, cex = 1.4)
text(sd(Rm1), mean(Rm1), "S&P 500", pos = 1, offset = 0.55, cex = 0.7,
     font = 2, col = c_spx)

for (nm in names(tr_E))
  points(tr_sd[nm], tr_E[nm], pch = 21, bg = pal[nm], col = surface,
        lwd = 1.6, cex = 1.5)
pt(tr_sd["Multigroup, SS"], tr_E["Multigroup, SS"], pal["Multigroup, SS"],
   "multigroup", 3, off = 0.7)

legend("bottomright", bty = "n", cex = 0.66, text.col = muted, ncol = 1,
      pch = 21, pt.bg = pal[names(tr_E)], col = surface, pt.lwd = 1.4,
      legend = names(tr_E))

ax(1, xtick, sprintf("%.0f%%", xtick * 100))
ax(2, pretty(yr), sprintf("%.0f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(a) The multigroup portfolio alongside every earlier portfolio", 3,
      line = 1.8, adj = 0, col = ink, cex = 1.02, font = 2)
mtext("Training period, historical covariance matrix. Blue curve is Project 2's frontier.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.7)
dev.off()

# ---- (b1) growth of one dollar over the testing period -----

open_png("fig_growth.png", h = 5.2)
gd <- c(dat2[1] - 31, dat2)
yr <- range(growth) * c(0.96, 1.06)
plot(NA, xlim = range(gd), ylim = yr, axes = FALSE, xlab = "", ylab = "")
yt <- pretty(yr)
abline(h = yt, col = grid, lwd = 1)
abline(h = 1, col = "#c3c2b7", lwd = 1.2, lty = 2)

for (nm in labels)
  lines(gd, growth[, nm], col = pal[nm], lwd = if (nm == "S&P 500") 2.8 else 1.7)

ax(2, yt, sprintf("%.1f", yt))
xt <- seq(as.Date("2022-01-01"), as.Date("2026-07-01"), by = "year")
axis(1, at = xt, labels = format(xt, "%Y"), col = "#c3c2b7", col.axis = muted,
     cex.axis = 0.75, lwd = 1)
legend("topleft", bty = "n", cex = 0.66, text.col = muted,
      lwd = 2, col = pal[labels], legend = labels)
mtext("Value of $1 invested", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(b1) Performance over the testing period", 3, line = 1.8, adj = 0,
      col = ink, cex = 1.02, font = 2)
mtext(sprintf("%s to %s, weights fixed at their training period values.",

```

```

        format(dat2[1], "%b %Y"), format(dat2[length(dat2)], "%b %Y")),
        3, line = 0.5, adj = 0, col = muted, cex = 0.7)
dev.off()

# ---- (b4) Fama decomposition on the return against beta plot -----

open_png("fig_fama.png", h = 5.2)
sml <- function(b) Rf + b * (mu_m - Rf)
xr <- c(0, max(beta_eq, 1) * 1.22); yr <- c(0, max(f_mu, sml(xr[2])) * 1.12)
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = pretty(xr), col = grid, lwd = 1)

abline(a = Rf, b = mu_m - Rf, col = c_front, lwd = 2.2)
text(xr[2], sml(xr[2]), "SML", pos = 3, offset = 0.3, cex = 0.75, font = 2,
     col = c_front)

segments(f_beta, sml(f_beta), f_beta, f_mu, col = c_eq, lwd = 6)
segments(beta_eq, sml(f_beta), beta_eq, sml(beta_eq), col = "#c9a227", lwd = 6)
segments(beta_eq, sml(beta_eq), beta_eq, f_mu, col = "#8a5cd6", lwd = 6)
segments(f_beta, f_mu, beta_eq, f_mu, col = "#c9c8c0", lwd = 1, lty = 2)
segments(f_beta, sml(f_beta), beta_eq, sml(f_beta), col = "#c9c8c0", lwd = 1, lty = 2)

pt(f_beta, f_mu, "#b8431a", "portfolio", 2)
points(f_beta, sml(f_beta), pch = 21, bg = surface, col = c_front, lwd = 2, cex = 1.2)
points(beta_eq, sml(beta_eq), pch = 21, bg = surface, col = c_front, lwd = 2, cex = 1.2)
pt(1, mu_m, c_spx, "market", 4, off = 0.5)

text(f_beta, (sml(f_beta) + f_mu) / 2, "selectivity", pos = 4, offset = 0.5,
     cex = 0.7, font = 2, col = c_eq)
text(beta_eq, (sml(f_beta) + sml(beta_eq)) / 2, "diversification", pos = 4,
     offset = 0.5, cex = 0.7, font = 2, col = "#c9a227")
text(beta_eq, (sml(beta_eq) + f_mu) / 2, "net selectivity", pos = 4, offset = 0.5,
     cex = 0.7, font = 2, col = "#8a5cd6")

ax(1, pretty(xr), sprintf("%.1f", pretty(xr)))
ax(2, pretty(yr), sprintf("%.1f%", pretty(yr) * 100))
mtext(expression(paste("Beta  ", beta[p], " and ", sigma[p]/sigma[m], " ")),
       1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(b4) Fama decomposition, single index with no short sales", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.02, font = 2)
mtext(sprintf("Left tick is beta = %.3f; right tick is sd_p/sd_m = %.3f.",
              f_beta, beta_eq), 3, line = 0.5, adj = 0, col = muted, cex = 0.7)
dev.off()

cat("wrote fig_multigroup.png, fig_growth.png, fig_fama.png\n")

```